

Unit 2 Guided Notes — Exponents & Scientific Notation

Same base? Move the exponents. Big or tiny number? Pack it into a $\times 10^n$.

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

This packet weaves three parts together: **Guided Notes** with worked examples, **Practice** built into each section to try as you learn, and a **Cumulative Review** with a full **Answer Key** at the end. Fill in the blanks and show your work on your printed copy.

Topics Covered: 2.1 Exponent Rules — Product, Quotient, Power, Zero, Negative (NC.8.EE.1) · 2.2 Scientific Notation — Format & Conversions (NC.8.EE.3) · 2.2 Multiplying & Dividing Scientific Notation (NC.8.EE.4) · 2.2 Word Problems with Scientific Notation (NC.8.EE.4)

⚠ **Parentheses come first!** A power attaches only to what it touches. $-3^2 = -9$, but $(-3)^2 = 9$. And a power outside parentheses hits EVERYTHING inside: $(3x^2)^3 = 27x^6$ (the 3 is cubed too). Keep an eye on parentheses all unit long.

1 Product & Quotient Rules

➤ **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Fill the examples and key phrases on your printed copy.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	_____	_____	_____
Quotient Rule	$x^a \div x^b = x^{a-b}$	_____	_____	_____

PRODUCT RULE: COUNT THE FACTORS

$$\begin{array}{c} \boxed{x} \boxed{x} \boxed{x} \\ x^3 \text{ (3 factors)} \end{array} \times \begin{array}{c} \boxed{x} \boxed{x} \\ x^2 \text{ (2)} \end{array} = \begin{array}{c} \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} \\ x^5 \text{ (5 factors)} \end{array}$$

$$x^3 \times x^2 = x^{3+2} = x^5 \rightarrow \text{ADD the exponents}$$

See WHY

$$\text{Product Rule} \cdot x^3 \times x^2 = x^5$$

$$\text{See why} \rightarrow \boxed{x} \boxed{x} \boxed{x} \times \boxed{x} \boxed{x} = \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} = \boxed{x^5}$$

Count the factors: $3 + 2 = 5$. Same base $\rightarrow$ ADD the exponents.

$$\text{Quotient Rule} \cdot x^5 \div x^2 = x^3$$

$$\text{See why} \rightarrow \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} \div \boxed{x} \boxed{x} \rightarrow \text{cancel 2 pairs} \rightarrow \boxed{x} \boxed{x} \boxed{x} = \boxed{x^3}$$

Cancel matching factors: $5 - 2 = 3$. Same base $\rightarrow$ SUBTRACT the exponents.

Quick Check

MC Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?

☐ Product Rule $\rightarrow x^8$ ☐ Power Rule $\rightarrow x^{15}$ ☐ Quotient Rule $\rightarrow x^2$

Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 — Multiply coefficients: $4 \times 3 = 12$

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

PRACTICE 1 Product & Quotient

Rules: Multiply $\rightarrow$ ADD exponents · Divide $\rightarrow$ SUBTRACT. Write each quotient as a reduced fraction. **Show your work in the box.**

1

Simplify $x^4 \cdot x^7$

2

Simplify $(3x^2)(5x^6)$

3

Simplify $(2a^3b^2)(4a^2b^5)$

4

Simplify $(-3x^4y)(6xy^3)$

5

Simplify $(5m^2n)(2mn^3)(3n^2)$

6

Simplify $\frac{x^9}{x^4}$

7

Simplify $\frac{15a^7}{3a^2}$

8

Simplify $\frac{24x^5y^6}{6x^2y^2}$

9

Simplify $\frac{20m^8n^3}{5m^3n}$

10

Simplify $\frac{-18p^6q^5}{3p^2q^2}$

11

Simplify $\frac{(2x^3)(6x^4)}{4x^2}$

12

Simplify $\frac{(3a^2b)(4a^3b^2)}{6ab}$

2 Power Rules

A power outside parentheses applies to every factor inside. Fill in, then check.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Power Rule	$(x^a)^b = x^{a \cdot b}$	_____	_____	_____
Power of a Product	$(xy)^n = x^n y^n$	_____	_____	_____
Power of a Quotient	$(\frac{x}{y})^n = \frac{x^n}{y^n}$	_____	_____	_____

⚠ PARENTHESES MATTER!

$-3^2 = \underline{\hspace{2cm}}$ vs. $(-3)^2 = \underline{\hspace{2cm}}$

Without (), ONLY the base gets the exponent — not the negative sign!

See WHY

Power Rule • $(x^2)^3 = x^6$

See why → $x^2 \cdot x^2 \cdot x^2 = x \cdot x \cdot x \cdot x \cdot x \cdot x = x^6$

Three copies of x^2 : $2 \times 3 = 6$. Power to a power → MULTIPLY.

ERROR ANALYSIS Spot the Error — Simplify $(3x^2)^3$. A student wrote $3x^6$.

The coefficient 3 must ALSO be cubed: $3^3 = 27$. Correct: $27x^6$. The power rule applies to EVERYTHING inside the parentheses.

Worked Example — Simplify $(2a^3b)^4$

Step 1 — Power the coefficient: $2^4 = 16$

Step 2 — Multiply the a exponent: $a^{3 \times 4} = a^{12}$

Step 3 — Multiply the b exponent: $b^{1 \times 4} = b^4$

✓ **Step 4 — Combine:** $16a^{12}b^4$

PRACTICE 2 Power Rules

Rule: Power to a power → MULTIPLY; apply the power to EVERY factor, top and bottom.

1

Simplify $(x^4)^5$

2

Simplify $(2x^3)^4$

3

Simplify $(-3a^2)^3$

4

Simplify $(x^2y^5)^3$

5

Simplify $(4m^3n^2)^2$

6

Simplify $(2x^3y^4)^5$

7

Simplify $(\frac{x^2}{y^3})^4$

8

Simplify $(\frac{2a}{b^2})^3$

9

Simplify $(\frac{3x^4}{2y})^2$

10

Simplify $((x^3)^2)^4$

11

Simplify $(2x^2y)^3(3xy^2)^2$

12

Simplify $(-2a^3b^2)^4$

3 Zero & Negative Exponents

Anything to the 0 power is 1; a negative exponent flips to a fraction.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Zero Exponent	$x^0 = 1$	_____	_____	_____
Negative Exponent	$x^{-n} = \frac{1}{x^n}$	_____	_____	_____

See WHY

Zero Exponent · $x^0 = 1$

See why → $\boxed{x} \boxed{x} \boxed{x} \div \boxed{x} \boxed{x} \boxed{x} = 1$, but by the quotient rule = $x^{3-3} = x^0$. So $x^0 = 1$

Anything (except 0) divided by itself is 1 → $x^0 = 1$.

Negative Exponent · $x^{-3} = \frac{1}{x^3}$

See why → $\boxed{x} \boxed{x} \div \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} = x^{2-5} = x^{-3}$; cancelling leaves $\frac{1}{x^3}$

A negative exponent means 'flip it' → $x^{-n} = \frac{1}{x^n}$.

Match the Rule to its Key Phrase

MATCH Draw a line from each rule to its key phrase.

Rule

Product Rule

Quotient Rule

Power Rule

Negative Exponent

Key Phrase

Power to power? MULTIPLY

Same base? ADD exponents

Flip to a fraction

Same base? SUBTRACT exponents

Negative Exponents Across the Fraction Bar

A negative exponent moves its factor to the OTHER side of the bar and turns positive.

Move across the bar: $x^{-3} = \frac{1}{x^3} \cdot \frac{1}{x^{-3}} = x^3 \cdot \frac{x^{-2}}{y^{-4}} = \frac{y^4}{x^2}$

In parentheses — flip to the reciprocal: $(\frac{a}{b})^{-n} = (\frac{b}{a})^n$. $(\frac{2}{3})^{-2} = (\frac{3}{2})^2 = \frac{9}{4} \cdot (\frac{x^2}{y})^{-3} = (\frac{y}{x^2})^3 = \frac{y^3}{x^6}$

PRACTICE 3 Zero & Negative Exponents

Write every answer with POSITIVE exponents only, as a reduced fraction.

1

Simplify $(7x^4y^2)^0$

2

Simplify $(3a^2b)^0$

3

Evaluate 4^{-3}

4

Positive exponents: $2x^{-5}$

5Simplify $\frac{x^{-3}}{y^{-2}}$ **6**Simplify $\frac{a^{-2}b^3}{c^{-1}}$ **7**Simplify $\frac{12m^{-2}n^4}{3m^4n^{-1}}$ **8**Simplify $\frac{8x^{-3}y^2}{2xy^{-1}}$ **9**Simplify $(\frac{x}{y})^{-2}$ **10**Evaluate $(\frac{2}{3})^{-3}$ **11**Simplify $(\frac{a^2}{b})^{-3}$ **12**Simplify $(\frac{3x}{y^2})^{-2}$

2 Scientific Notation

$$a \times 10^n \quad \text{where} \quad 1 \leq a < 10$$

4.5×10^7 | coefficient · base (always 10) · exponent

SCIENTIFIC NOTATION: MOVE THE DECIMAL

$$45,000 \rightarrow 4.5 \times 10^4$$

big number → move LEFT 4 places → exponent +4

$$0.0067 \rightarrow 6.7 \times 10^{-3}$$

small number → move RIGHT 3 places → exponent -3

Type	Direction	Exponent	Example
Large number	Move decimal LEFT	_____	$45,000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT	_____	$0.0067 = 6.7 \times 10^{-3}$

PRACTICE B Scientific Notation Conversions

Remember: $1 \leq a < 10$. Positive exp = BIG · Negative exp = SMALL.

1

45,000,000 → sci. notation

2

0.00032 → sci. notation

3

6.02×10^8 → standard

4

9.1×10^{-5} → standard

5

780,000 → sci. notation

6

0.0000067 → sci. notation

7

3.45×10^4 → standard

8

1.2×10^{-3} → standard

9

5,280,000,000 → sci. notation

10 2.5×10^{-2} → standard
11

0.000000904 → sci. notation

12 7.15×10^6 → standard
13

63,400,000,000 → sci. notation

14 8.8×10^{-6} → standard

Multiplying Scientific Notation — 3-Step Process

1. _____ the coefficients (the a-values)
2. _____ the exponents
3. Adjust if coefficient is NOT between _____ and _____

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients: $3 \times 2 = 6$

ADD exponents: $4 + 5 = 9$

✓ **Answer:** 6×10^9

Dividing Scientific Notation — 3-Step Process

1. _____ the coefficients
2. _____ the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

Divide coefficients: $8 \div 4 = 2$

SUBTRACT exponents: $6 - 2 = 4$

✓ **Answer:** 2×10^4

⚠ ADJUSTMENT RULES:

- If coefficient ≥ 10 : Move decimal LEFT, ADD 1 to exponent
- If coefficient < 1 : Move decimal RIGHT, SUBTRACT 1 from exponent

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

Multiply: $5 \times 4 = 20$

Add exponents: $3 + 2 = 5 \rightarrow$ result: 20×10^5

Adjust ($20 \geq 10$): $20 = 2.0 \times 10^1$, so ADD 1 to exponent

✓ **Final answer:** 2.0×10^6

ERROR ANALYSIS **Spot the Error — Convert 0.0045 to scientific notation.** A student wrote 4.5×10^3 .

Small number $\rightarrow$ NEGATIVE exponent. Correct: 4.5×10^{-3} . Numbers less than 1 always have negative exponents in scientific notation.

PRACTICE C Operations with Scientific Notation

Multiply/divide coefficients · add/subtract exponents · adjust so $1 \leq a < 10$.

1

$$(3 \times 10^4)(2 \times 10^5)$$

2

$$(8 \times 10^7) \div (4 \times 10^3)$$

3

$$(5 \times 10^3)(7 \times 10^6)$$

4

$$(9.6 \times 10^8) \div (3.2 \times 10^5)$$

5

$$(6 \times 10^5)(5 \times 10^4)$$

6

$$(1.5 \times 10^{-3})(4 \times 10^7)$$

7

$$(7.2 \times 10^9) \div (1.8 \times 10^{-2})$$

8

$$(2.5 \times 10^6)(8 \times 10^{-3})$$

Word Problems with Scientific Notation

Strategy: Identify what operation to use. Write your scientific-notation setup. Adjust if needed.

PRACTICE D Word Problems

Identify the operation, write your setup, then adjust so $1 \leq a < 10$.

1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds for light to reach Earth?

Answer: _____

2. A bacteria cell is about 2×10^{-6} m long; a virus about 5×10^{-8} m. How many times longer is the bacteria?

Answer: _____

3. The US debt is about 3.4×10^{13} dollars; population about 3.4×10^8 . What is the debt per person?

Answer: _____

4. A grain of sand has mass 6.5×10^{-4} g. A beach has 8×10^{18} grains. Total mass in grams?

Answer: _____

5. A body has 3.7×10^{13} cells, each 1×10^{-5} m wide. Lined up, how long in meters?

Answer: _____

6. A computer does 4.8×10^9 calculations per second. A task needs 1.44×10^{12} . How many seconds?

Answer: _____

7. A hard drive holds 2×10^{12} bytes. Each photo is 4×10^6 bytes. How many photos fit?

Answer: _____

8. Light travels 3×10^8 m/s. How far in 6×10^3 seconds?

Answer: _____

★ Cumulative Practice — Unit 2 Review ★

Mixed practice over everything above. Show all work; circle your final answer.

REVIEW · A Exponent Rules

1

Simplify $x^4 \cdot x^6$

2

Simplify $(3x^2)(4x^5)$

3

Simplify $(2a^3b^2)(5ab^4)$

4

Simplify $\frac{24x^7}{6x^3}$

5

Simplify $(m^3)^5$

6

Simplify $(2x^2y^3)^4$

7

Simplify $\frac{18a^5b^6}{3a^2b^2}$

8

Simplify $(\frac{x^4}{y^2})^3$

REVIEW · B Scientific Notation Conversions

1

7,200,000 → sci. notation

2

0.00045 → sci. notation

3

3.05×10^7 → standard

4

6.1×10^{-5} → standard

5

91,000 → sci. notation

6

0.0000008 → sci. notation

7

 $4.44 \times 10^6 \rightarrow$ standard

8

 $2.7 \times 10^{-3} \rightarrow$ standard
REVIEW · C Operations

1

 $(3 \times 10^4)(3 \times 10^5)$

2

 $(8 \times 10^9) \div (2 \times 10^3)$

3

 $(5 \times 10^6)(4 \times 10^3)$

4

 $(9.6 \times 10^8) \div (1.6 \times 10^2)$

5

 $(7 \times 10^5)(8 \times 10^2)$

6

 $(1.2 \times 10^{-3})(4 \times 10^5)$

REVIEW · D Word Problems

1. A red blood cell is about 8×10^{-6} m wide; a platelet about 2×10^{-6} m. How many times wider is the red blood cell?

Answer: _____

2. Earth is about 1.5×10^8 km from the Sun. A spacecraft travels 3×10^4 km/h. How many hours for the trip?

Answer: _____

3. A factory makes 5×10^3 widgets per day. How many in 2×10^2 days?

Answer: _____

4. A virus is about 3×10^{-7} m and a cell about 6×10^{-5} m. How many times larger is the cell?

Answer: _____

REVIEW · E Negative Exponents & Rational Expressions

Write every answer with positive exponents only, as a reduced fraction.

1

Evaluate 5^{-2}

2

Positive exponents: $3x^{-4}$

3

Simplify $\frac{x^{-5}}{y^{-3}}$

4

Simplify $(\frac{a}{b})^{-3}$

5

Simplify $\frac{10a^{-2}b^5}{2a^3b^{-1}}$

6

Simplify $(\frac{2x^2}{y})^{-2}$

★ Answer Key ★

Do not distribute to students until after the lesson.

Guided Notes — Fill-Ins

§1 Product & Quotient. Product: $x^3 \times x^4 = x^7$; $(2x^3)(3x^2) = 6x^5$; “Same base? ADD.” Quotient: $x^6 \div x^2 = x^4$; $(12x^7) \div (3x^2) = 4x^5$; “Same base? SUBTRACT.”

§2 Power. Power: $(x^2)^3 = x^6$; $(2x^3)^2 = 4x^6$; “Power to power? MULTIPLY.” Power of a Product: $(2x)^3 = 8x^3$; $(3ab)^2 = 9a^2b^2$; “Give EACH factor the power.” Power of a Quotient: $(2/3)^2 = \frac{4}{9}$; $(a/b)^3 = \frac{a^3}{b^3}$; “Power to top AND bottom.” Parentheses: $-3^2 = -9$ vs $(-3)^2 = 9$.

§3 Zero & Negative. Zero: $5^0 = 1$; $(3x^5)^0 = 1$; “ANY base⁰ = 1.” Negative: $2^{-3} = \frac{1}{8}$; $4x^{-3} = \frac{4}{x^3}$; “Flip to a fraction.” **Across the bar:** $\frac{x^{-2}}{y^{-4}} = \frac{y^4}{x^2}$.

Reciprocal: $(\frac{x^2}{y})^{-3} = (\frac{y}{x^2})^3 = \frac{y^3}{x^6}$.

2.2 Scientific Notation. Large $\rightarrow$ +exp; Small $\rightarrow$ -exp. Multiply: MULTIPLY coefficients, ADD exponents, adjust to 1-10. Divide: DIVIDE coefficients, SUBTRACT exponents. Examples: $(3 \times 10^4)(2 \times 10^5) = 6 \times 10^9$; $(8 \times 10^6) \div (4 \times 10^2) = 2 \times 10^4$; $(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 \rightarrow 2.0 \times 10^6$.

Practice 1 — Product & Quotient

1) x^{11} 2) $15x^8$ 3) $8a^5b^7$ 4) $-18x^5y^4$ 5) $30m^3n^6$ 6) x^5 7) $5a^5$ 8) $4x^3y^4$ 9) $4m^5n^2$ 10) $-6p^4q^3$ 11) $3x^5$ 12) $2a^4b^2$

Practice 2 — Power Rules

1) x^{20} 2) $16x^{12}$ 3) $-27a^6$ 4) x^6y^{15} 5) $16m^6n^4$ 6) $32x^{15}y^{20}$ 7) $\frac{x^8}{y^{12}}$ 8) $\frac{8a^3}{b^6}$ 9) $\frac{9x^8}{4y^2}$ 10) x^{24} 11) $72x^8y^7$ 12) $16a^{12}b^8$

Practice 3 — Zero & Negative

1) 1 2) 1 3) $\frac{1}{64}$ 4) $\frac{2}{x^5}$ 5) $\frac{y^2}{x^3}$ 6) $\frac{b^3c}{a^2}$ 7) $\frac{4n^5}{m^6}$ 8) $\frac{4y^3}{x^4}$ 9) $\frac{y^2}{x^2}$ 10) $\frac{27}{8}$ 11) $\frac{b^3}{a^6}$ 12) $\frac{y^4}{9x^2}$

Practice B — Conversions

1) 4.5×10^7 2) 3.2×10^{-4} 3) 602,000,000 4) 0.000091 5) 7.8×10^5 6) 6.7×10^{-6} 7) 34,500 8) 0.0012 9) 5.28×10^9 10) 0.025 11) 9.04×10^{-7} 12) 7,150,000 13) 6.34×10^{10} 14) 0.0000088

Practice C — Operations

1) 6×10^9 2) 2×10^4 3) 3.5×10^{10} 4) 3×10^3 5) 3.0×10^{10} 6) 6×10^4 7) 4×10^{11} 8) 2.0×10^4

Practice D — Word Problems

1) $(9.3 \times 10^7) \div (1.86 \times 10^5) = 5 \times 10^2$ s (500 s) 2) 40× longer 3) $1 \times 10^5 = \$100,000$ 4) 5.2×10^{15} g 5) 3.7×10^8 m 6) 3×10^2 s (300 s) 7) $(2 \times 10^{12}) \div (4 \times 10^6) = 5 \times 10^5$ photos 8) $(3 \times 10^8)(6 \times 10^3) = 1.8 \times 10^{12}$ m

Cumulative Review

A • Exponent Rules: 1) x^{10} 2) $12x^7$ 3) $10a^4b^6$ 4) $4x^4$ 5) m^{15} 6) $16x^8y^{12}$ 7) $6a^3b^4$ 8) $\frac{x^{12}}{y^6}$

B • Conversions: 1) 7.2×10^6 2) 4.5×10^{-4} 3) 30,500,000 4) 0.000061 5) 9.1×10^4 6) 8×10^{-7} 7) 4,440,000 8) 0.0027

C • Operations: 1) 9×10^9 2) 4×10^6 3) 2×10^{10} 4) 6×10^6 5) 5.6×10^8 6) 4.8×10^2

D • Word Problems: 1) 4× wider 2) 5×10^3 h (5,000 h) 3) 1×10^6 widgets 4) $(6 \times 10^{-5}) \div (3 \times 10^{-7}) = 2 \times 10^2 = 200 \times$ larger

E • Negative Exponents: 1) $\frac{1}{25}$ 2) $\frac{3}{x^4}$ 3) $\frac{y^3}{x^5}$ 4) $\frac{b^3}{a^3}$ 5) $\frac{5b^6}{a^5}$ 6) $\frac{y^2}{4x^4}$